

TAHIA ANAN DHIRA

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EDUCATION

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|---|------|
| <ul style="list-style-type: none"> • MSc in Applied Social Data Science (ASDS) (Degree: Distinction)
The London School of Economics and Political Science, London, United Kingdom
Relevant coursework: Social Network Analysis; Quantitative Text Analysis; Applied Machine Learning; Managing and Visualizing Data; Research Design | 2023 |
| <ul style="list-style-type: none"> • MSS in Economics (CGPA: 3.88 / 4.00)
University of Dhaka, Dhaka, Bangladesh | 2017 |
| <ul style="list-style-type: none"> • BSS in Economics (CGPA: 3.94 / 4.00)
University of Dhaka, Dhaka, Bangladesh | 2016 |

RESEARCH INTERESTS

Computational Social Science, Environmental Economics, Economic and Social Networks, Applied Econometrics, Machine Learning for Social Science, Quantitative Text Analysis

RESEARCH MANUSCRIPTS

Rethinking the Income-Emissions Relationship: Evidence from Trade-Embodied CO₂ Emissions Networks (*working paper, 2026*)

- Developed reproducible computational workflows for integrating and processing large-scale international datasets on embodied carbon emissions, trade, and macroeconomic indicators.
- Constructed weighted global trade networks and implemented longitudinal network analyses to characterize the evolution of global embodied carbon trade networks.
- Integrated network analysis and panel econometrics to examine how economic development and network position shape domestic and trade-embodied CO₂ emissions across advanced and developing economies.

Global migration network and the geography of development (*working paper, 2026*)

- Created a longitudinal panel of directed, weighted bilateral migration networks spanning **23 years** to examine global migration dynamics.
- Developed reproducible computational workflows for network construction, visualization, and longitudinal analysis using R.
- Contributed to network model development, core-periphery detection, interpretation of network statistics, and manuscript preparation.

Gendered Cultural Hierarchies in Social Media Audience Discourse: Evidence from Bangladeshi Artists (*working paper, 2026*)

- Developed scalable Python pipelines using the YouTube Data API v3 for automated collection, preprocessing, and analysis of **over 8 million multilingual social media comments**.
- Applied computational text mining and natural language processing techniques, including sentiment analysis (VADER) and structural topic modelling (quanteda/STM), to identify patterns in audience discourse and sentiment across artists, genders, and artistic categories.
- Developed a human-in-the-loop classification workflow for a large-scale YouTube video dataset, using the Gemini API for preliminary classification of artist gender and artistic category from video titles, followed by manual validation and correction of approximately 48,000 videos.
- Built reproducible computational pipelines integrating automated data collection, preprocessing, statistical modelling, visualization, and documentation using Python and R.

Product Reproducibility and International Trade Patterns: Evidence from Cultural Goods (*under review, 2026*)

- Constructed computational pipelines for harmonizing, processing, and analyzing large-scale international trade data covering 190 countries.
- Combined structural gravity modelling (PPML) with dynamic network modelling (STERGM) to examine the formation and persistence of international trade relationships.
- Developed reproducible research workflows integrating data processing, statistical modelling, visualization, version control, and documentation using Python, R, and Git.
- Prepared and submitted the manuscript as the corresponding author.

RESEARCH EXPERIENCE

Capstone Project, MS ASDS, LSE | 2023

Formation of friendship network: Analysing assortment of pro-sociality and personality type

- Designed and implemented a behavioural classroom experiment to study homophily and friendship network formation.
- Estimated Exponential Random Graph Models (ERGMs) to identify structural and attribute-based drivers of network formation.
- Examined the role of personality traits, and prosocial behaviour in network formation

Coursework: ST445, MS ASDS, LSE | 2022–2023

Topic-based content and sentiment analysis of COVID-19 vaccine: a study using Reddit

- Contributed to the manuscript preparation and network visualization.

Mental Health of University Students Before and After Online Education

University Grants Commission (UGC)-funded project | 2020–2021

- Contributed to the design and implementation of a longitudinal survey examining changes in university students' mental health before and after the transition to online education.
- Coordinated data collection and contributed to survey-instrument validation and quantitative analysis.

Willingness to Pay for Treating Mental vs. Physical Illness

Bangladesh Medical Research Council (BMRC)-funded project | 2020–2021

- Contributed to the coordination and implementation of a comparative survey for university students and adults.

PUBLICATIONS

- Huque, R., **Dhira, T. A.**, et al. (2023). Determinants of client satisfaction with health insurance: Evidence using a mixed-method approach from Bangladesh. *SSM - Health Systems*, 3. <https://doi.org/10.1016/J.SSMHS.2025.100058>
- Rahman, M. A., **Dhira, T. A.**, et al. (2022). PHQ-9 validation in Bangladesh. *PLOS ONE*.
<https://doi.org/10.1371/journal.pone.0269634>
- Dhira, T. A.**, Rahman, M. A., et al. (2021). GAD-7 validation in Bangladesh. *PLOS ONE*.
<https://doi.org/10.1371/journal.pone.0261590>
- Mehareen, J., Rahman, M. A., **Dhira, T. A.**, et al. (2021). Mental health during COVID-19. *Journal of Affective Disorders Reports*. <https://doi.org/10.1016/j.jadr.2021.100179>

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Economics, University of Dhaka	October 2025 – present
Lecturer, Department of Economics, University of Dhaka	November 2018 – September 2025
- Taught 10 different undergraduate courses in economics. - Supervised undergraduate dissertations and research projects. - Trained STATA to the undergraduate and graduate students. - Developed computational course materials using R, Python, and Stata.	
Lecturer, Department of Economics, East West University	March 2018 – October 2018

COMPUTATIONAL & QUANTITATIVE SKILLS

Programming Languages

- Python (pandas, NumPy, NetworkX, Matplotlib),
- R (igraph, statnet, ergm, tergm, stm, networkDynamic, quanteda),
- STATA

Network Analysis

Directed and weighted networks; dynamic networks; centrality; community detection; core-periphery analysis; ERGM; STERGM

Econometrics

Panel econometrics; PCSE; FGLS; PPML/structural gravity; nonlinear and interaction models

Machine Learning & NLP

Applied machine learning; sentiment analysis (VADER); structural topic modelling (STM); quantitative text analysis

Research Workflow & Tools

Git/GitHub; Gemini API; reproducible workflows; data processing; visualization; computational debugging and validation; LaTeX

Survey/Data Collection: Qualtrics; YouTube Data API

AWARDS AND SCHOLARSHIP

Commonwealth Master's Scholarship	2022
Multiple Gold Medals for highest CGPA (BSS & MSS Economics)	2016, 2017

LANGUAGE

Advanced fluency in English and Bangla